NAT-ICS Analytical Reasoning: One-Day Intensive Preparation Guide

Prof. Bashir

1. Purpose of This Guide

This document is designed for one-day preparation of the Analytical Reasoning section of the NAT-ICS test.

The Analytical Reasoning section usually tests logical thinking through:

- scenario-based reasoning,
- statement-based reasoning,
- arrangements,
- grouping,
- ranking,
- syllogisms,
- assumptions and conclusions.

This section does not require memorization. It requires careful reading, diagram-making, and elimination of wrong options.

2. Core Rules for Analytical Reasoning

2.1. Rule 1: Do Not Assume Extra Information

Only use the information given in the question. Do not use personal knowledge or real-world assumptions.

Example:

> Statement: Some students are players.

You cannot conclude that all students are players. You also cannot conclude that all players are students.

2.2. Rule 2: Convert Words into Symbols

Use symbols to save time.

Symbol	Meaning
$>$	greater than, taller than, ahead of, older than
$<$	less than, shorter than, behind, younger than
$=$	same group, same position, same category
$\neq$	not equal, not together, not same
$\rightarrow$	if this happens, then that happens
$\therefore$	therefore

2.3. Rule 3: Fixed Conditions First

Always identify fixed conditions before flexible conditions.

Fixed condition example:

> A is at the left end.

Flexible condition example:

> B is somewhere to the right of C.

The fixed condition should be placed first in your diagram.

2.4. Rule 4: Use Elimination

In MCQs, it is often faster to remove impossible choices than to directly find the answer.

3. Important Logical Vocabulary

Term	Meaning
Must be true	It is definitely correct according to the given information.
Could be true	It is possible, but not guaranteed.
Cannot be true	It violates at least one given condition.
Only if	Shows a necessary condition.
If	Shows a sufficient condition.
Either/or	One of the two options must occur.
Neither/nor	Both options are rejected.
Some	At least one. It does not mean many or most.
All	Every member of the group.
No	Zero overlap between groups.

4. Part 1: Linear Arrangement

4.1. Basic Idea

Linear arrangement questions involve people or objects placed in a row.

Common words:

- left,
- right,

- before,
- after,
- between,
- immediately before,
- immediately after,
- at either end.

4.2. Important Difference

Phrase	Meaning
A is to the left of B	A is somewhere on the left side of B. Not necessarily next to B.
A is immediately left of B	A is directly next to B on the left side.
A is between B and C	A is placed somewhere between B and C.
A is exactly between B and C	Equal number of spaces exist on both sides of A.

4.3. Solved Example 1

Five students A, B, C, D, and E are standing in a row.

- A is to the left of B.
- C is to the right of B.
- D is not at either end.
- E is at the left end.

Question

Who is at the left end?

(A) A

(B) B

(C) C

(D) E

Answer: D
Explanation: The statement directly says that E is at the left end.

4.4. Solved Example 2

Six people P, Q, R, S, T, and U are sitting in a row.

- P is immediately left of Q.
- R is to the right of Q.

- S is at the left end.
- T is not next to S.
- U is to the right of R.

Question

Which pair must sit together?

(A) P and Q

(B) Q and R

(C) R and U

(D) S and T

Answer: A
Explanation: The phrase “immediately left” means P must be directly beside Q.

5. Part 2: Ranking and Order

5.1. Basic Idea

Ranking questions compare people or things by height, age, marks, speed, weight, or position. Use symbols:

$$A > B$$

This means A is greater than B. Depending on the question, it can mean taller, older, faster, or higher-ranked.

5.2. Solved Example

Ali is taller than Bilal. Bilal is taller than Daniyal. Daniyal is taller than Hamza.

$$Ali > Bilal > Daniyal > Hamza$$

Question

Who is the shortest?

(A) Ali

(B) Bilal

(C) Daniyal

(D) Hamza

Answer: D
Explanation: Hamza is at the end of the order, so he is the shortest.

5.3. Common Trap

If the question asks who is second tallest, do not select the first name. Read the exact position required.

6. Part 3: Grouping and Selection

6.1. Basic Idea

Grouping questions divide people or objects into teams, rooms, subjects, days, or categories.
Important words:

- with,
- without,
- together,
- not together,
- exactly,
- at least,
- at most,
- only one,
- neither.

6.2. Solved Example

Six students A, B, C, D, E, and F are divided into two groups of three.

- A and B cannot be in the same group.
- C must be with D.
- E cannot be with F.

Question

If A is in Group 1 and C is in Group 1, who else must be in Group 1?

(A) B

(B) D

(C) E

(D) F

Answer: B
Explanation: C must be with D. Since C is in Group 1, D must also be in Group 1.

7. Part 4: Scheduling Questions

7.1. Basic Idea

Scheduling questions involve assigning activities to days, periods, rooms, or times.

Make a table.

Day	Activity
Monday	
Tuesday	
Wednesday	
Thursday	
Friday	

7.2. Solved Example

Five tests English, Math, Physics, Computer, and Urdu are scheduled from Monday to Friday.

- Math is before Physics.
- English is on Monday.
- Computer is after Physics.
- Urdu is not on Friday.

Question

Which test is on Monday?

(A) English

(B) Math

(C) Physics

(D) Urdu

Answer: A
Explanation: The condition directly says that English is on Monday.

8. Part 5: Statement and Conclusion

8.1. Basic Idea

A conclusion follows only if it is definitely supported by the given statement.

8.2. Do Not Use Outside Knowledge

Example:

Statement: All engineers are educated.

Valid conclusion:

Some educated people are engineers.

Invalid conclusion:

All educated people are engineers.

8.3. Solved Example

Statement: Some books are novels. All novels are interesting.

Conclusion I: Some books are interesting. Conclusion II: All books are interesting.

(A) Only I follows

(B) Only II follows

(C) Both follow

(D) Neither follows

Answer: A

Explanation: Some books are novels, and all novels are interesting. Therefore, some books are interesting. But all books are not necessarily interesting.

9. Part 6: Syllogism

9.1. Basic Forms

Statement	**Meaning**
All A are B	Every A is inside B.
Some A are B	At least one A is B.
No A is B	A and B do not overlap.
Some A are not B	At least one A is outside B.

9.2. Rules

1. "All A are B" does not mean "All B are A."
2. "Some A are B" means at least one overlap.
3. "No A is B" means complete separation.
4. If all A are B and all B are C, then all A are C.
5. If some A are B and all B are C, then some A are C.

9.3. Solved Example

Statements:

- All cats are animals.
- All animals are living things.

Conclusion:

- All cats are living things.

Answer: Conclusion follows

Explanation: Cats are inside animals, and animals are inside living things. So cats are also inside living things.

10. Part 7: Statement and Assumption

10.1. Basic Idea

An assumption is an unstated idea that must be accepted for the statement to make sense.

10.2. Solved Example

Statement: The college advised students to reach the exam center one hour early.

Assumption I: Students may face delay on the way. Assumption II: The exam will be cancelled.

(A) Only I is implicit

(B) Only II is implicit

(C) Both are implicit

(D) Neither is implicit

Answer: A

Explanation: The advice makes sense if delays are possible. Nothing suggests that the exam will be cancelled.

11. Part 8: Strong and Weak Arguments

11.1. Basic Idea

A strong argument is directly related, practical, and based on logic. A weak argument is emotional, irrelevant, exaggerated, or unsupported.

11.2. Solved Example

Question: Should students practice past papers before the NAT test?

Argument I: Yes, because past papers help students understand the pattern and time pressure. Argument II: No, because studying is boring.

(A) Only I is strong

(B) Only II is strong

(C) Both are strong

(D) Neither is strong

Answer: A

Explanation: Argument I is logical and relevant. Argument II is personal and weak.

12. Part 9: Cause and Effect

12.1. Basic Idea

You must decide whether one event caused another, both are effects of a common cause, or there is no clear relation.

12.2. Solved Example

Statements:

- Many students failed the exam this year.
- The paper was much more difficult than usual.

Possible relation:

(A) First is the cause, second is the effect.

(B) Second is the cause, first is the effect.

(C) Both are unrelated.

(D) Both are effects of another cause.

Answer: B
Explanation: A difficult paper can cause many students to fail.

13. Fast Solving Methods

13.1. Method 1: Table Method

Use this for grouping and scheduling.

Person	**Group**	**Restriction**
A		Not with B
B		Not with A
C		With D
D		With C

13.2. Method 2: Line Method

Use this for ranking and order.

$$A > B > C > D$$

13.3. Method 3: Box Method

Use this for seating arrangement.

$$\boxed{1} \quad \boxed{2} \quad \boxed{3} \quad \boxed{4} \quad \boxed{5}$$

13.4. Method 4: Elimination Method

Check each option against the conditions. If an option breaks one condition, remove it.

14. Common Mistakes

1. Confusing "left of" with "immediately left of."
2. Reversing syllogism statements.
3. Assuming "some" means "most."
4. Ignoring words like not, only, except, must, cannot.
5. Spending too much time on one puzzle.
6. Solving without drawing a diagram.
7. Selecting an option that is possible when the question asks what must be true.

15. Practice Set 1: Basic Questions

Question 1

A is taller than B. B is taller than C. C is taller than D. Who is the shortest?

(A) A

(B) B

(C) C

(D) D

Answer: D

Question 2

Five people A, B, C, D, and E are sitting in a row. A is at the left end. B is immediately right of A. C is right of B. Who is immediately right of A?

(A) B

(B) C

(C) D

(D) E

Answer: A

Question 3

Statement: All pens are tools. Some tools are expensive.
Which conclusion follows?

(A) All pens are expensive.

(B) Some pens are expensive.

(C) Some tools are pens.

(D) No pen is a tool.

Answer: C

Question 4

Statement: No birds are mammals. All sparrows are birds.
Which conclusion follows?

(A) No sparrow is a mammal.

(B) All mammals are sparrows.

(C) Some mammals are birds.

(D) Some birds are mammals.

Answer: A

Question 5

If X is the brother of Y and Y is the sister of Z, then X is related to Z as:

(A) Father

(B) Brother

(C) Cousin

(D) Uncle

Answer: B

16. Practice Set 2: Scenario-Based Questions

Read the following scenario and answer the questions.
Five students Ali, Bilal, Hamza, Daniyal, and Usman are standing in a line.

- Ali is ahead of Bilal.
- Hamza is behind Bilal.
- Daniyal is ahead of Ali.
- Usman is behind Hamza.

The order is:

$$Daniyal > Ali > Bilal > Hamza > Usman$$

Question 1

Who is first in the line?

(A) Ali

(B) Bilal

(C) Daniyal

(D) Usman

Answer: C

Question 2

Who is immediately behind Bilal?

(A) Ali

(B) Hamza

(C) Daniyal

(D) Usman

Answer: B

Question 3

Who is last in the line?

(A) Usman

(B) Hamza

(C) Bilal

(D) Ali

Answer: A

17. Practice Set 3: Grouping Questions

Six candidates A, B, C, D, E, and F are assigned to two rooms, Room 1 and Room 2. Each room has three candidates.

- A cannot be with B.
- C must be with D.
- E cannot be with F.
- A is in Room 1.

Question 1

If C is in Room 1, who else must be in Room 1?

(A) B

(B) D

(C) E

(D) F

Answer: B

Question 2

If A, C, and D are in Room 1, which pair is in Room 2?

(A) B and E

(B) B and F

(C) E and F

(D) C and D

Answer: A or B
Explanation: Room 2 contains B plus E and F cannot be together. However, this setup is incomplete unless one more condition is provided. If only A, C, and D are fixed in Room 1, then Room 2 must contain B, E, and F, but this violates the condition that E cannot be with F. Therefore, A, C, and D cannot all be in Room 1.

18. Practice Set 4: Statement-Based Questions

Question 1

Statement: Some teachers are writers. All writers are creative.
Conclusion I: Some teachers are creative. Conclusion II: All teachers are creative.

(A) Only I follows

(B) Only II follows

(C) Both follow

(D) Neither follows

Answer: A

Question 2

Statement: All mobiles are devices. Some devices are costly.
Conclusion I: Some mobiles are costly. Conclusion II: Some devices are mobiles.

(A) Only I follows

(B) Only II follows

(C) Both follow

(D) Neither follows

Answer: B

Question 3

Statement: No honest person cheats. Ali is honest.
Conclusion: Ali does not cheat.

(A) Follows

(B) Does not follow

(C) Cannot be determined

(D) Invalid statement

Answer: A

19. Practice Set 5: Mixed NAT-Style MCQs

1. If all roses are flowers and some flowers fade quickly, which conclusion must follow?

   (A) All roses fade quickly.

   (B) Some roses fade quickly.

   (C) Some flowers are roses.

   (D) No roses are flowers.

   Answer: C

2. A is older than B. C is younger than B. D is older than A. Who is the oldest?

   (A) A

   (B) B

   (C) C

   (D) D

   Answer: D

3. Statement: The school increased library hours so students can study after classes.

   Assumption:

   (A) Students may need extra study time.

   (B) Students dislike books.

   (C) Teachers do not teach properly.

   (D) Exams are cancelled.

   Answer: A

4. If P is immediately before Q and Q is immediately before R, then:

   (A) R is before P.

   (B) P, Q, R are consecutive.

   (C) Q is after R.

   (D) P is after Q.

   Answer: B

5. Statement: No laptops are phones. Some phones are expensive.

   Which conclusion follows?

   (A) Some laptops are expensive.

   (B) No phone is a laptop.

   (C) All phones are laptops.

   (D) Some expensive things are laptops.

   Answer: B

6. Question: Should students manage time during NAT preparation?

   Argument I: Yes, because NAT has limited time and multiple sections. Argument II: No, because time management is unnecessary in tests.

   (A) Only I is strong

   (B) Only II is strong

   (C) Both are strong

   (D) Neither is strong

   Answer: A

7. Five books A, B, C, D, and E are placed on a shelf. A is immediately left of B. C is at the right end. D is not next to C. Which pair must be together?

   (A) A and B

(B) B and C

(C) C and D

(D) D and E

Answer: A

8. If some doctors are teachers and all teachers are educated, then:

(A) Some doctors are educated.

(B) All doctors are educated.

(C) No doctor is educated.

(D) All educated people are doctors.

Answer: A

9. If A is to the left of B and B is to the left of C, then:

(A) C is to the left of A.

(B) A is to the left of C.

(C) B is to the right of C.

(D) A is immediately left of C.

Answer: B

10. Statement: The city installed more street lights to reduce accidents at night.
Assumption:

(A) Poor lighting may cause accidents.

(B) Accidents only happen in daylight.

(C) Street lights cause accidents.

(D) People do not travel at night.

Answer: A

20. Answer Key

Practice Set 1

Question	**Answer**
1	D
2	A
3	C
4	A
5	B

Practice Set 2

Question	Answer
1	C
2	B
3	A

Practice Set 4

Question	Answer
1	A
2	B
3	A

Practice Set 5

Question	Answer
1	C
2	D
3	A
4	B
5	B
6	A
7	A
8	A
9	B
10	A

21. Final Revision Sheet

Arrangement

- “Immediately left” means directly next to the person.
- “Left of” means anywhere on the left side.
- Always draw boxes for seats or positions.

Ranking

- Convert every sentence into $>$ or $<$.
- Arrange from highest to lowest.
- Check whether the question asks highest, lowest, second highest, or middle.

Grouping

- Mark people who must be together.
- Mark people who cannot be together.
- Check each option against all conditions.

Syllogism

- All A are B does not mean all B are A.
- Some means at least one.
- No means zero overlap.
- Do not make conclusions beyond the statements.

Assumption

- An assumption is hidden but necessary.
- Avoid extreme assumptions.
- Choose the option that directly supports the statement.

22. Test-Day Strategy

1. Attempt direct questions first.
2. Skip long puzzles temporarily if they take too much time.
3. Draw small diagrams for arrangements and grouping.
4. Read negative words carefully: not, except, cannot, least, false.
5. Do not spend more than two minutes on one analytical question.
6. If stuck, eliminate impossible options and make the most logical choice.

23. One-Day Minimum Target

Before the test, make sure you can solve:

- 10 linear arrangement questions,
- 10 ranking questions,
- 10 grouping questions,
- 10 syllogism questions,
- 10 statement-conclusion questions,
- 10 assumption/argument questions.

This gives enough exposure to the main Analytical Reasoning patterns likely to appear in NAT-style tests.